Aneesh Komanduri

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Education

University of Arkansas — Fayetteville, Arkansas

Doctor of Philosophy (Ph.D.), Computer Science — **Aug 2021 - May 2026**

Master of Science (M.S.), Computer Science — **2021 - 2024**

Bachelor of Science (B.S.), Computer Science & Mathematics — **2017 - 2021**

Skills

- **Programming Languages:** Python, C/C++, Java, Javascript, SQL
- **Data & Cloud**: Apache Spark, Databricks, PostgreSQL, AWS
- **AI/ML**: PyTorch, TensorFlow, scikit-learn, Pyro, Hugging Face Transformers
- **Generative AI**: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LLM Agents, Parameter-Efficient Supervised Fine-Tuning (LoRA), GRPO, Diffusion Models
- **ML Systems**: DeepSpeed, Accelerate, Multi-GPU Distributed Computing, vLLM
- **Application Development**: Flask, Django, LangChain, MCP, A2A

Professional Experience

University of Colorado Anschutz School of Medicine — Aurora, Colorado

NLP Research Scientist — **June 2026 - Present**

- Developed novel paradigm of **mechanistic uncertainty quantification in LLMs** formulated as **causal robustness** to **activation steers** for more reliable/interpretable model predictions
- Built an efficient **multi-agent LLM orchestration framework** integrating specialized agents and lightweight information-theoretic **compression mechanisms** trained via **GRPO**
- Designing **scalable experimentation** and **evaluation pipelines** for assessing LLM reliability across models, datasets, and inference strategies in clinical applications

NEC Laboratories America, Inc. — Princeton, New Jersey

Research Scientist Intern — **May 2025 - Aug 2025**

- Designed core components of an **enterprise LLM-based AI assistant** for multi-aspect IT service requests, integrating information retrieval, representation learning, and generative AI
- Developed a **weakly supervised contrastive learning** framework for domain-specific semantic embedding refinement, improving **retrieval augmented generation (RAG)** performance by 20% compared with baseline retrieval approaches
- Engineered and evaluated data-preprocessing pipelines using LLM-based summarization and robust RAG evaluation using **LLM-as-a-Judge** paradigms, including context relevance, groundedness, and answer relevance

Research Experience

University of Arkansas — Fayetteville, Arkansas

Graduate Research Assistant — **Oct 2021 - June 2026**

Trustworthy Vision-Language Models

- Engineered a robust **benchmarking framework** for **causal reasoning in LVLMs**; annotated three physical system image datasets, formulated three novel causal tasks, and benchmarked 9 models under **zero-shot, few-shot, and chain-of-thought prompting** [EMNLP'25]
- Established a realistic setting for **unlearning sensitive information in LVLMs** and designed a **resource-efficient, cross-attention guided** method that targets least-attended visual tokens, significantly improving unlearning-utility trade-offs compared to full fine-tuning baselines [LockLLM@NeurIPS'25]

Causal Generative Modeling

- Proposed and implemented theoretically novel and empirically robust **VAE framework** for **causal representation learning** on high-dimensional data, outperforming state-of-the-art baselines by 15%–20% in disentanglement metrics [IJCAI'24]
- Designed a novel interpretable **causally-aware diffusion model** for generating high-quality **image counterfactuals** that outperforms state-of-the-art baseline methods in the quality of the generated counterfactuals along both quantitative and qualitative metrics [ECAI'24]

Selected Publications

Manuscripts Under Review

- **Aneesh Komanduri**, Majid Afshar, Timothy A. Miller, and Yanjun Gao. **Measuring Mechanistic Uncertainty in Large Language Models via Causal Robustness**. *Under review*, 2026.
- He Cheng, Xin Chen, **Aneesh Komanduri**, Saksham Khatwani, Timothy A. Miller, Majid Afshar, and Yanjun Gao. **Information-Theoretic and Uncertainty-Guided Communication Compression in Multi-Agent Systems**. *Under review*, 2026.
- **Aneesh Komanduri** and Xintao Wu. **Leveraging Foundation Models for Causal Generative Modeling**. *Under review*, 2026.

Peer-Reviewed Publications

- Karuna Bhaila, **Aneesh Komanduri**, Minh-Hao Van, and Xintao Wu. **Cross-Modal Attention Guided Unlearning in Large Vision-Language Models**. *Lock-LLM Workshop at NeurIPS*, 2025.
- **Aneesh Komanduri**, Karuna Bhaila, and Xintao Wu. **CausalVLBench: Benchmarking Visual Causal Reasoning in Large Vision-Language Models**. *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- **Aneesh Komanduri**. **Toward Causal Generative Modeling: From Representation to Generation**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
- **Aneesh Komanduri**, Chen Zhao, Feng Chen, and Xintao Wu. **Causal Diffusion Autoencoders: Toward Counterfactual Generation via Diffusion Probabilistic Models**. *Proceedings of the European Conference on Artificial Intelligence (ECAI)*, 2024.
- **Aneesh Komanduri**, Yongkai Wu, Feng Chen, and Xintao Wu. **Learning Causally Disentangled Representations via the Principle of Independent Causal Mechanisms**. *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
- **Aneesh Komanduri**, Xintao Wu, Yongkai Wu, and Feng Chen. **From Identifiable Causal Representations to Controllable Counterfactual Generation: A Survey on Causal Generative Modeling**. *Transactions on Machine Learning Research (TMLR)*, 2024.

Honors and Awards

Silver Reviewer Award — May. 2026
Awarded by International Conference on Machine Learning (ICML)

1st Place Graduate Student Poster @ NSF 2024 DART Conference ($1,500) — Sep. 2024
Awarded by Arkansas Economic Development Commission (AEDC)

Doctoral Academy Fellowship ($48,000) — 2021-2025
Awarded by University of Arkansas Graduate School and International Education

Service

Reviewer: AAAI'27, NeurIPS'26, ICML'26, AAAI'26, ECAI'25, KDD'25, PAKDD'25, IJCAI'25, ICML'25, ICLR'25, NeurIPS'24, LoG'24, ICMLA'24, TMLR, DMLR, IJDSA, Pattern Recognition